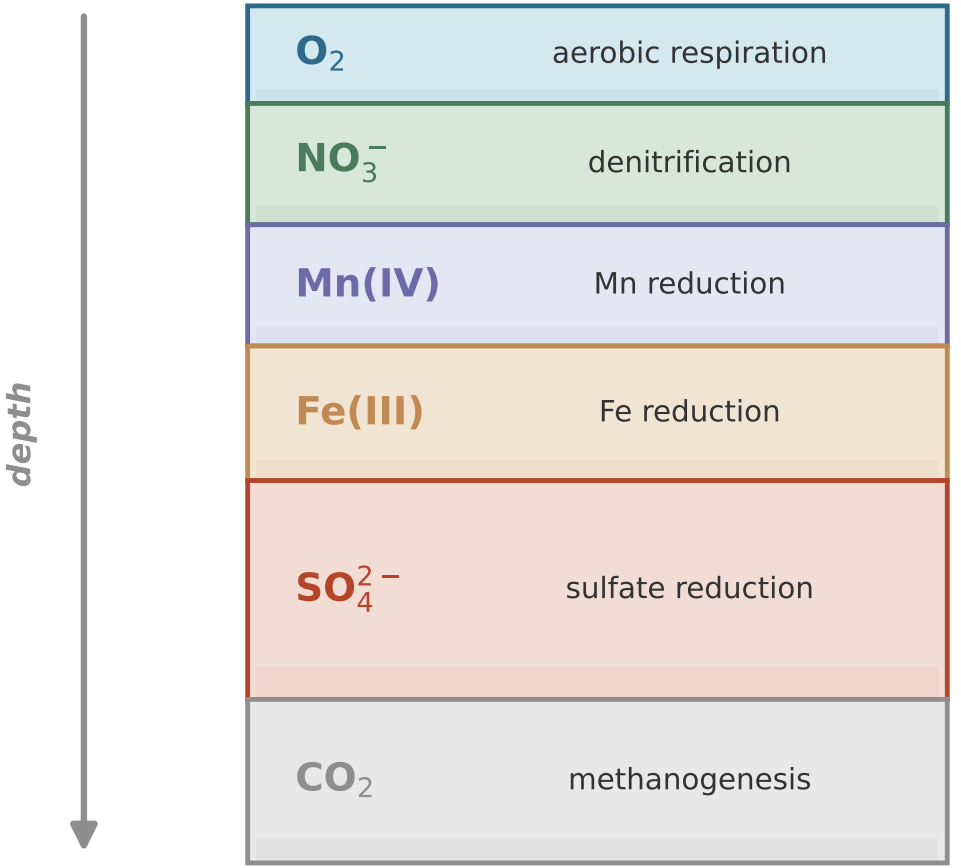The terminal electron acceptor hierarchy